2.0 Projection of Points (Manual Drawing)

SEMESTER: I/II

COURSE TITLE: **COMPUTER AIDED ENGINEERING GRAPHICS**

COURSE CODE: **22MED13/23**

Solution Manual

Prepared by :

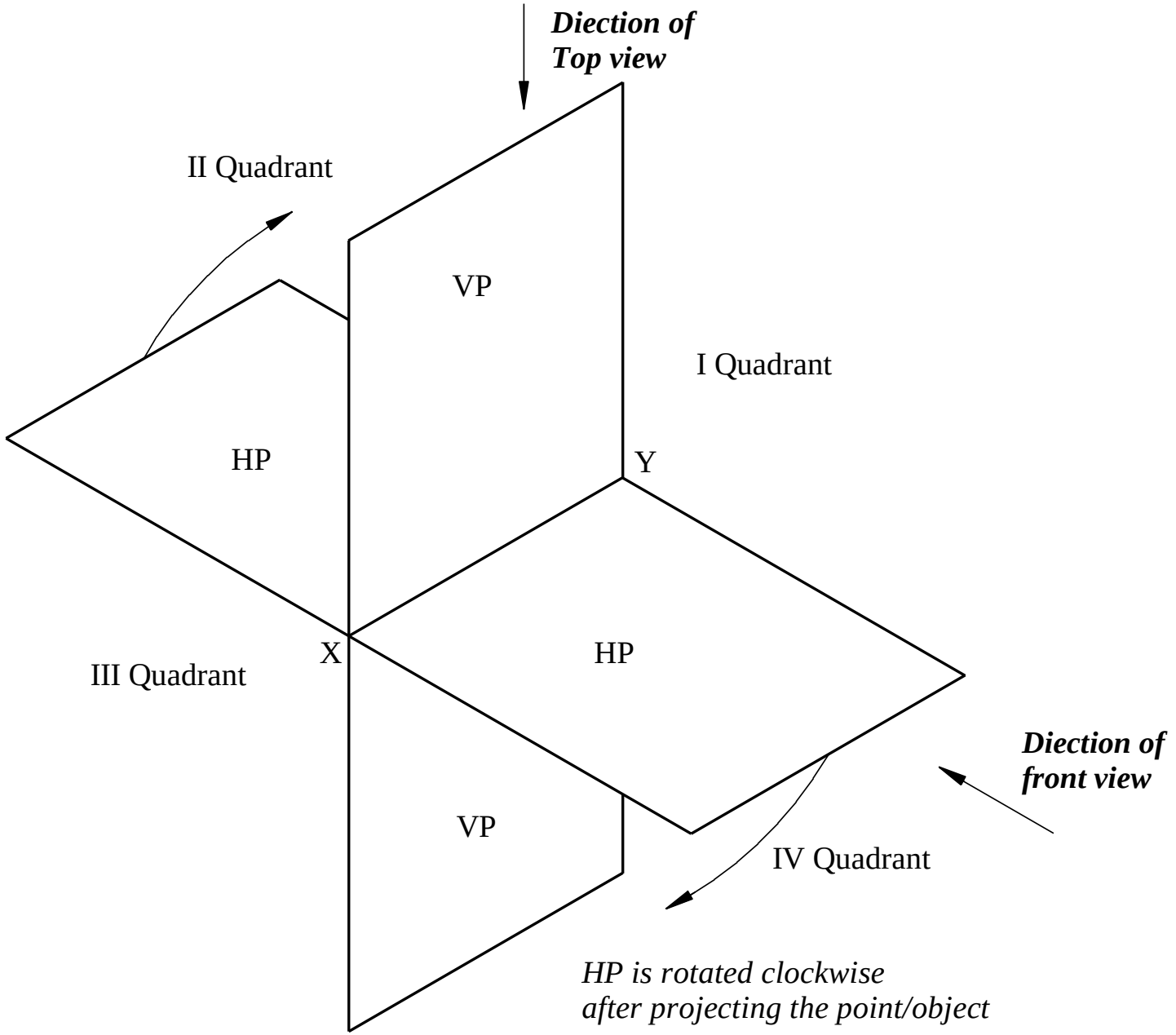
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Projection of Points in 4 quadrants

- A point in I quadrant will be above HP & in front of VP and in front of LPP/RPP.
- A point in II Quadrant will be above HP and behind VP.
- A point in III quadrant will be below HP, behind VP and behind LPP/RPP.
- A point in the IV quadrant will be below HP, in front of VP.
- In I quadrant, ***(First angle Projection)*** as object comes first and then the plane, right side view will be on LPP & vice versa.
- In III quadrant, ***(Third angle Projection)*** plane comes first and then object, hence right side will be on RPP & left side view on LPP.

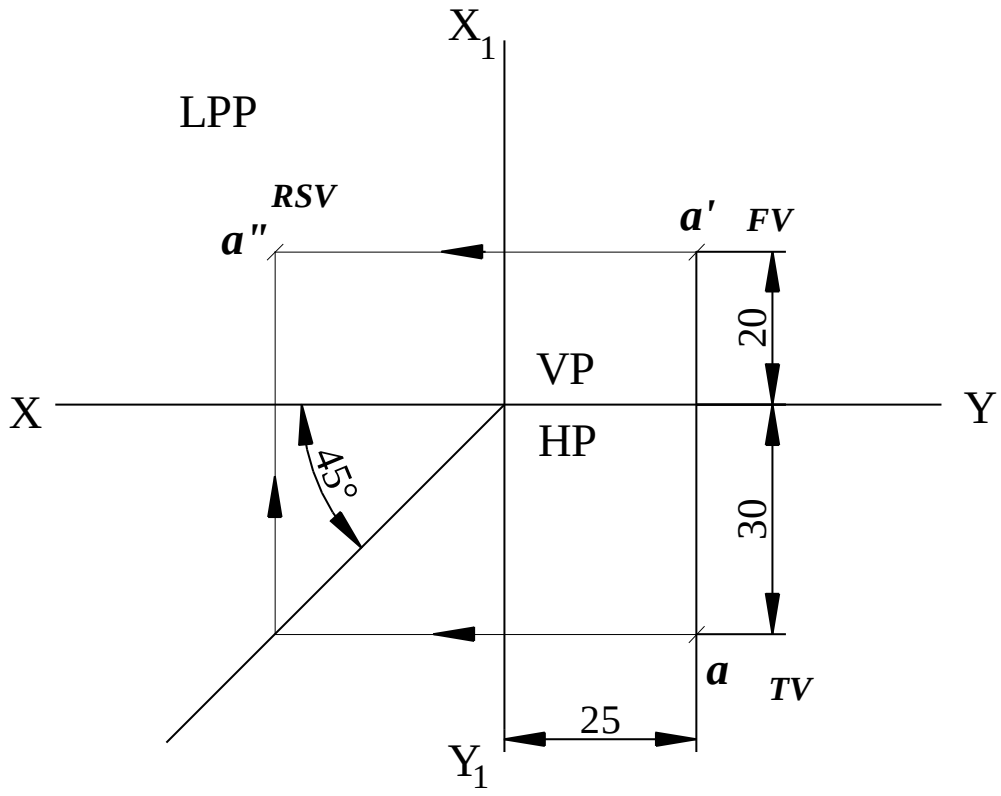


2.1 Point A is 30mm in front of VP, 20mm above HP and 25mm in front of LPP. Draw the projections.

Solution: The point A is in I quadrant. Hence, VP will be above XY line & HP will be below XY line.

- Draw front view ***a'*** 20 mm above XY line and top view ***a*** 30 mm below XY line (along the same projector)
- To obtain the side view, draw X_1Y_1 line 25 mm towards **left** (As LPP is given) of projector ***aa'*** . From the intersection of XY and X_1Y_1 line, draw a 45^0 line wherever the top view is *(In this case, **below** XY line)*
- Draw a horizontal projector from top view to intercept the 45^0 line and then draw a vertical projector to meet the horizontal drawn from the front view.
- The intersection of the two lines gives side view ***a"*** ***(Right side view projected on LPP)***

Point A in is I Quadrant



Solution to Q 2.1

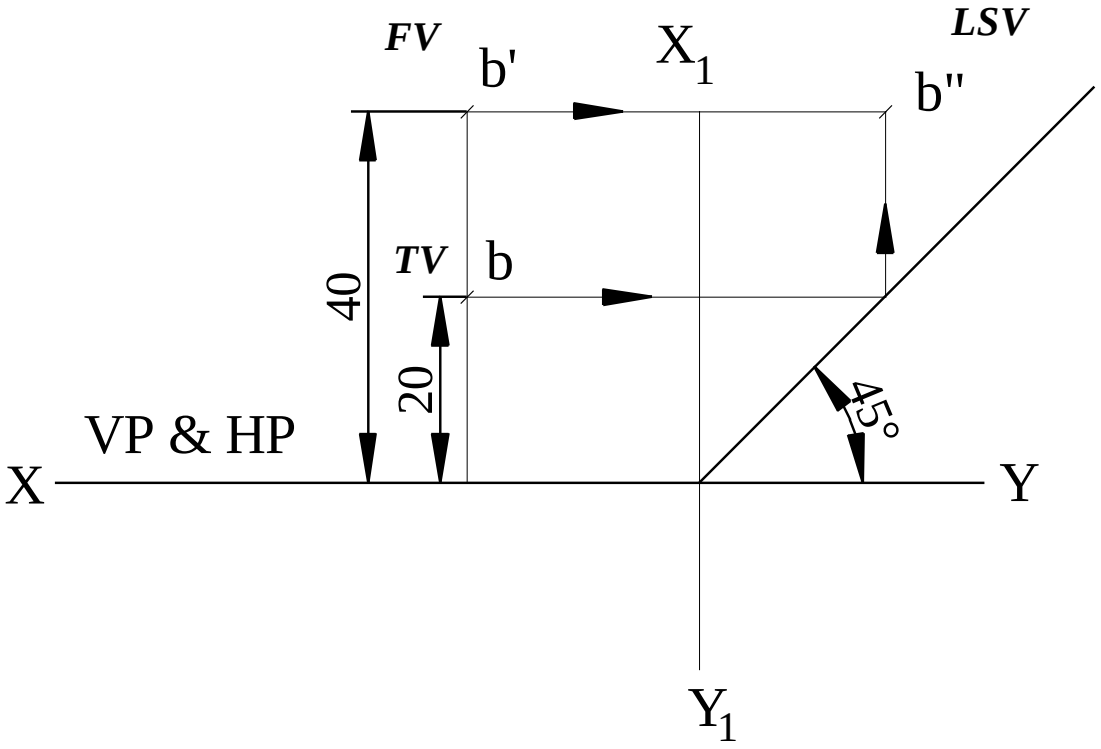
Note: *1. Above HP implies front view is above XY line, In front of VP implies top view is below XY line.*
2. In First angle projection, Left side view is projected on RPP & Right side view is projected on LPP.
3. The front view & side view of a point will be along the same horizontal line

2.2 Point B is 20mm behind VP, 40mm above HP and 25mm in front of RPP. Draw its projections.

Solution: The point B is in II quadrant. Hence, both HP & VP will be above XY line (after rotating HP in cw direction)

- Draw front view ***b'*** 40 mm above XY line and top view ***b*** 20 mm above XY line (along the same projector)
- To obtain the side view, draw X_1Y_1 line 25 mm towards **right** (As RPP is given) of projector ***bb'*** . From the intersection of XY and X_1Y_1 line, draw a 45^0 line wherever the top view is *(In this case, **above** XY line)*
- Draw a horizontal projector from top view to intercept the 45^0 line and then draw a vertical projector to meet the horizontal drawn from the front view.
- The intersection of the two lines gives side view ***b" (left view as it is mentioned in front of RPP implies observer is looking from left side)***

Point B in is II Quadrant



Solution to Q 2.2

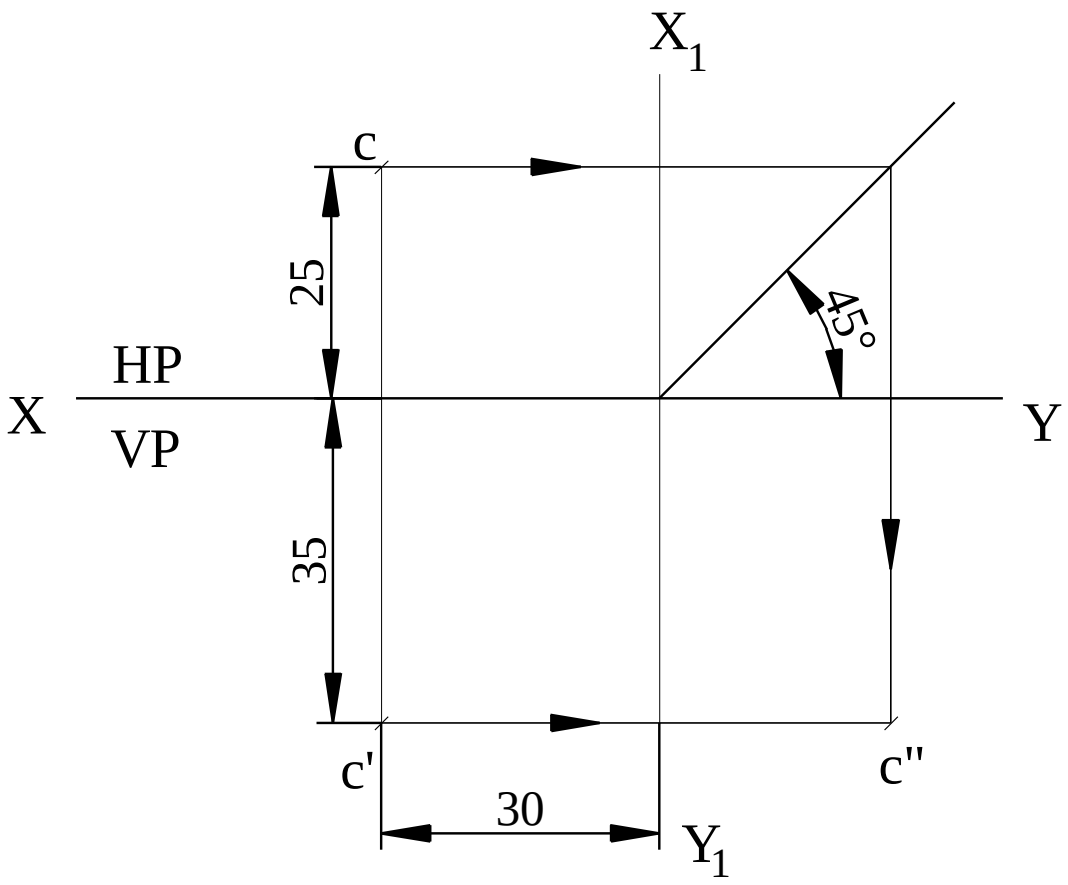
Note: *1. Above HP implies front view is above XY line, Behind VP implies top view is above XY line.*
2. The front view & side view of a point will be along the same horizontal line.

2.3 Point C is 25mm behind VP, 35mm below HP and 30mm behind RPP. Draw its projections.

Solution: The point C is in III quadrant. Hence, HP will be above XY line & VP will be below XY line (after rotating HP in cw direction)

- Draw front view ***c'*** 35 mm below XY line and top view ***c*** 25 mm above XY line (along the same projector)
- To obtain the side view, draw X_1Y_1 line 30 mm towards **right** (As RPP is given) of projector ***cc'*** . From the intersection of XY and X_1Y_1 line, draw a 45^0 line wherever the top view is *(In this case, **above** XY line)*
- Draw a horizontal projector from top view to intercept the 45^0 line and then draw a vertical projector to meet the horizontal drawn from the front view.
- The intersection of the two lines gives side view ***c"*** ***(Right view as in III angle, the LSV will be on LPP & RSV on RPP)***

Point C in is III Quadrant



Solution to Q 2.3

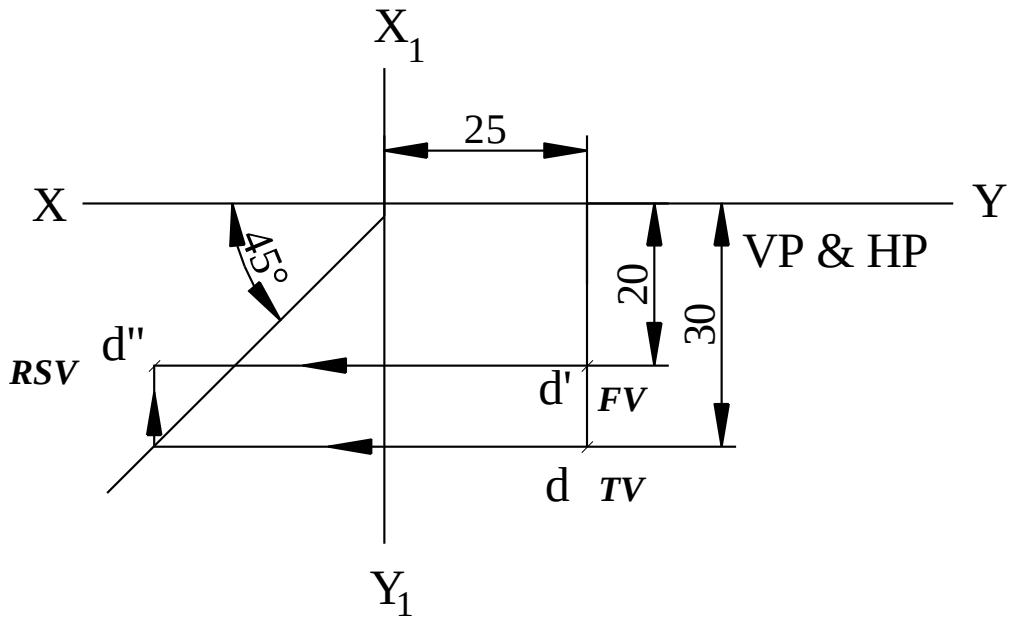
Note: *1. Below HP implies front view is below XY line, Behind VP implies top view is above XY line.*
2. The front view & side view of a point will be along the same horizontal line.

2.4 Point D is 30mm in front of VP, 20mm below HP and 25mm in front of LPP. Draw the projections.

Solution: The point D is in IV quadrant. Hence, both HP & VP will be below XY line. (after rotating HP in cw direction)

- Draw front view ***d'*** 20 mm below XY line and top view ***d*** 30 mm above XY line (along the same projector)
- To obtain the side view, draw X_1Y_1 line 25 mm towards **left** (as LPP is given) of projector ***dd'*** . From the intersection of XY and X_1Y_1 line, draw a 45^0 line wherever the top view is *(In this case, **below** XY line)*
- Draw a horizontal projector from top view to intercept the 45^0 line and then draw a vertical projector to meet the horizontal drawn from the front view.
- The intersection of the two lines gives side view ***d"*** ***(Right view as it is stated that point is in front of LPP)***

Point D in is IV Quadrant



Solution to Q 2.4

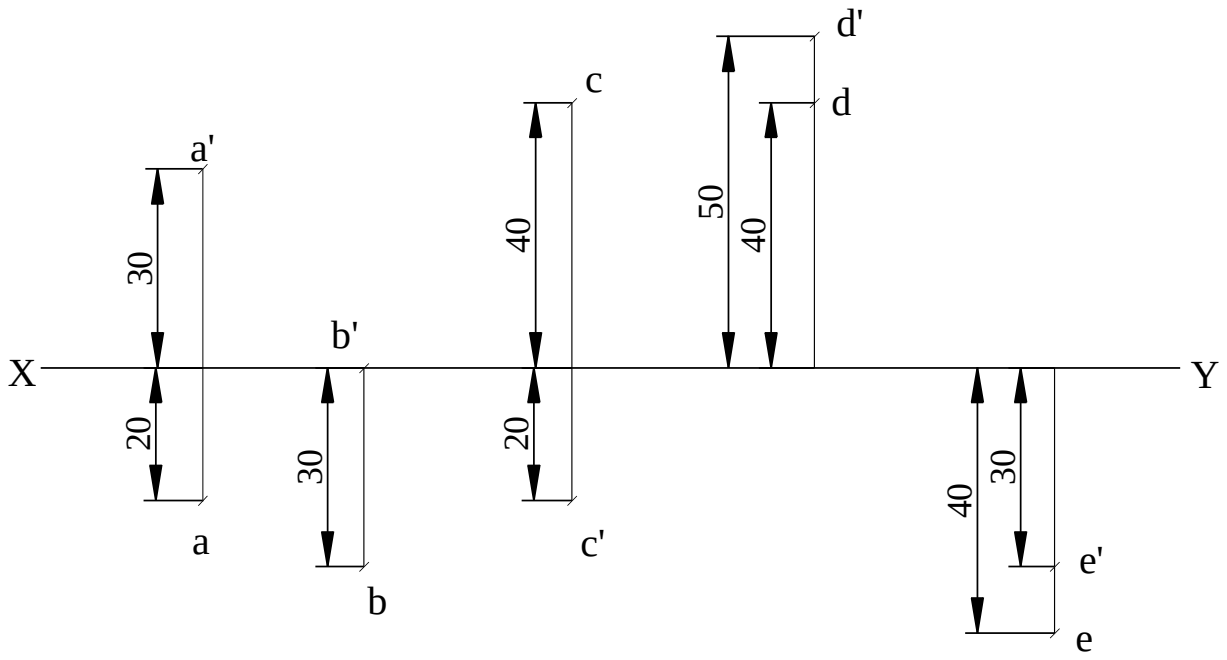
Note: *1. Below HP implies front view is below XY line, In front of VP implies top view is below XY line.*
2. The front view & side view of a point will be along the same horizontal line.

Important Points to be remembered

1. *When a Point is **on HP**, its front view is on XY line, & when a point is **in VP,** its top view will be on XY line.*
2. *When a point lies **on HP & in front of VP,** the point lies in either **I or IV** quadrant.*
3. *When a point lies **on HP and behind VP,** the point lies either **II or III** quadrant.*
4. *When a point lies **in VP and above HP,** the point is either in **I or II** quadrant.*
5. *When a point lies **in VP and below HP,** the point is either in **III or IV** quadrant*
6. *When point lies **in both HP & VP**, its front and top views coincide and will be on XY line.*
7. *When a point lies **in HP, VP & Profile plane**, all three views coincide & will be at intersection of XY & X_1Y_1 line.*

2.5 Draw the projections of the following points on the same XY line.

a) A is 20mm in front of VP and 30 mm above HP.
b) B is 30mm in front of VP and in HP.
c) C is 40mm behind VP and 20mm below HP
d) D is 40mm behind VP and 50mm above HP
e) E is 40mm in front of VP and 30mm below HP.



Solution to Q 2.5

Solution:

(a) Point A is in I Quadrant: Front view of A is 30 mm above XY line & its top view is 20 mm below XY line.

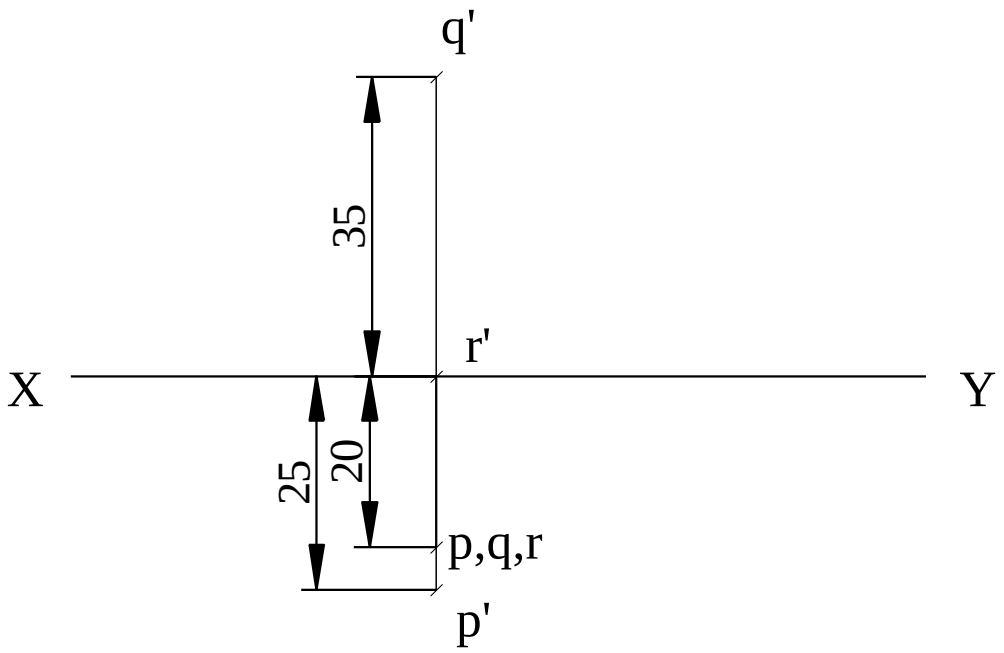
(b) Point B is either in I or IV quadrant: Front view of B is on XY & its top view is 30 mm below XY line.

(c) Point C is in III Quadrant: Front view of C is 20 mm above XY line & its top view is 40 mm above XY line.

(d) Point D is in II Quadrant: Front view of D is 50 mm above XY line & its top view is 40 mm above XY line.

(e) Point E is in IV Quadrant: Front view of E is 30 mm below XY line & its top view is 40 mm below XY line.

2.6 A point 20mm below XY line is the top view of three points P, Q and R. P is 25mm below HP, Q is 35mm above HP and R on HP. Draw the projections of the three points and state their positions with reference planes and the quadrants in which they lie.



Solution to Q 2.6

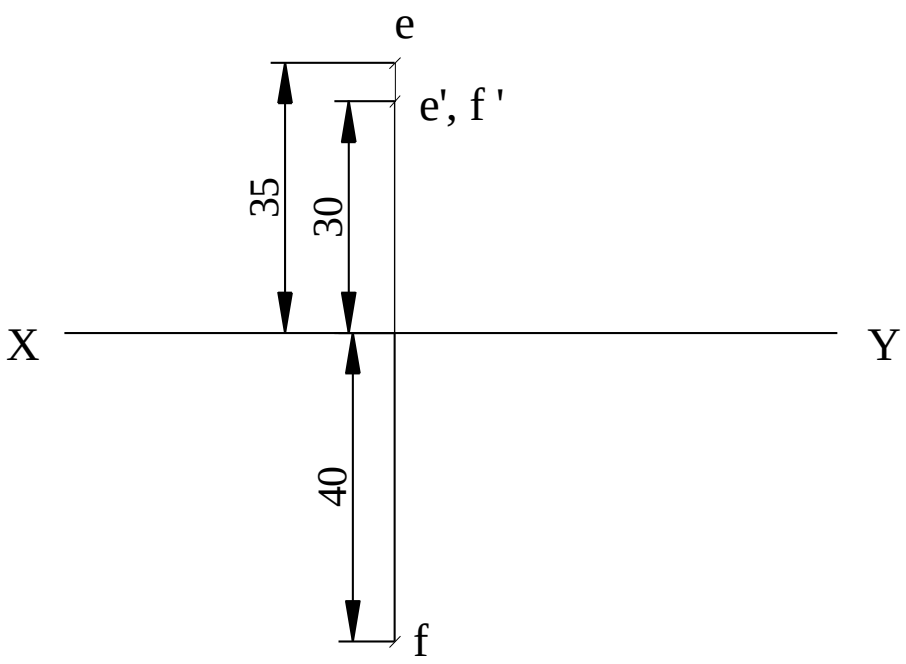
Solution:

Point P is in IV Quadrant: P is 25 mm below HP & 20 mm in front of VP.

Point Q is in I quadrant: Point Q is 35 mm above HP & 20 mm in front of VP.

Point R is either in I or IV quadrant: Point R is on HP & 20 mm in front of VP.

2.7 A point 30mm above XY line is the front view of two points E and F. E is 35mm behind VP and F is 40mm in front of VP. Draw the projections of the two points and state their positions with reference planes and the quadrants in which they lie.



Solution to Q 2.7

Solution:

Point E is in II Quadrant: E is 30 mm above HP & 35 mm behind VP.

Point F is in I quadrant: Point F is 30 mm above HP & 40 mm in front of VP.